Rediawan Ibrahim

Software Engineer | Full-Stack Developer

Jakarta, Indonesia | +62 896-9321-8083 | rediawanibrahim15@gmail.com

---

Professional Summary

Software Engineer and Full-Stack Developer with professional experience building web applications, backend services, and REST APIs using TypeScript, Angular, Go, and relational databases. Experienced across the software development lifecycle, including frontend development, database integration, testing, debugging, and workflow automation, with a focus on maintainable architecture, reliable systems, and practical business solutions.

Completed the Micro1 Frontier Engineering Challenge 2026 by building Agentic Repository Debugger, a reproducible 12-case benchmark for evaluating AI coding-agent repairs in a Go-based commerce API. Focused on automated evaluation, regression testing, reproducibility, and evidence-based validation. Interested in developer tools, AI evaluation systems, backend reliability, and internal platforms that solve complex engineering problems.

Education

Institut Bisnis Nusantara | Bachelor of Computer Systems | September 2017 – February 2022

Professional Experience

Universitas Indonesia Maju

Full-Stack Developer | May 2023 – August 2025

- Designed and developed responsive web applications using Angular, TypeScript, HTML, and SCSS, focusing on usability, maintainability, and frontend consistency.
- Built and maintained backend services and REST APIs to support data processing, system integration, and business workflows.
- Contributed to end-to-end application development, including frontend-backend integration, debugging, and ongoing application maintenance.

PT Reycom Document Solusi

IT Programmer | November 2022 – May 2023

- Developed desktop applications using C# and Microsoft SQL Server to support business operations and automate data processing workflows.
- Built an Excel-to-PDF automation system for processing large datasets and generating documents with less manual intervention.
- Implemented and maintained application functionality involving database operations, document generation, and data handling.

Projects

SentinelPay — Agentic Payment Incident Commander

Airwallex Agentic Banking Hackathon 2026 | Planned October 2026

- Planned an agentic payment-operations system to investigate payment failures, correlate transaction and account context, and identify likely root causes using Airwallex APIs in a controlled sandbox.
- Designed a bounded-autonomy workflow with policy-based action controls, human approval for sensitive operations, and explicit authorization boundaries.
- Defined a verification and escalation approach in which the system checks post-action state, records supporting evidence, and escalates incidents when recovery is unsafe or outcomes remain uncertain.

Agentic Repository Debugger

Micro1 Frontier Engineering Challenge 2026 | August 2026

- Built a reproducible 12-case benchmark in a Go-based commerce API to evaluate AI coding agents on realistic software defects, including concurrency, transaction handling, validation, pricing, persistence, and HTTP error mapping.
- Implemented 12 case-specific evaluators and Git-based bug-injection-to-repair provenance to trace changes and validate agent-generated fixes.
- Validated repairs through regression, repeated, and shuffled test execution, supported by reproduction instructions, baseline-versus-final patches, agent trajectories, an improvement changelog, and evidence for all 12 cases.

SignalForge

Personal Project | Planned

- Planned an engineering-output evaluation platform that assesses functional correctness, requirement compliance, testing, failure handling, documentation, evidence completeness, regression risk, and maintainability.
- Designed configurable evaluation rules and evidence-backed pass, warning, and fail classifications, with evaluation history and structured findings.
- Planned regression comparisons and human-review escalation to make quality decisions reproducible, traceable, and supported by observable evidence.

Failure-Driven Service Lab

Personal Project | Planned

- Planned a reproducible backend reliability environment covering database failures, timeouts, retry exhaustion, transaction rollback, concurrency issues, resource exhaustion, and graceful degradation.
- Designed a scenario-based workflow for failure injection, service execution, failure detection, recovery handling, and expected-behavior validation.
- Planned automated health checks, integration and regression tests, execution logs, and baseline-versus-resilient behavior comparisons to evaluate service reliability.

Bred AI Workspace

Personal Project | In Development (Currently Paused)

- Developed a full-stack AI-powered workspace concept focused on productivity and internal engineering workflows using React, TypeScript, Go, Fiber, and PostgreSQL.

- Worked on a modular application architecture incorporating REST APIs, authentication, state management, form validation, and database persistence.
- Focused on maintainable code, reusable components, clean architecture, and automated testing to support future development and extensibility.

Technical Skills

- **Frontend** **:** React, TypeScript, Vue.js, Angular, HTML, CSS, SCSS, Bootstrap
- **Backend** **:** Go, Fiber, Node.js, Express.js, C#, .NET, REST APIs
- **Database** **:** PostgreSQL, MySQL, Microsoft SQL Server
- **Tools** **:** Docker, Git, Cloudflare, Postman
- **Focus Area** **:** Automated Testing, Regression Testing, API Integration, Workflow Automation, AI Coding-Agent Evaluation